Traffic Light Controller

EE25Btech11063 (Yalamati Vejith),
EE25Btech11064 (Yojit manral)

February 20, 2026

1 Traffic Light Controller

1.1 Aim

The objective is to design and implement a Traffic Light Controller using JK flip-flops, ensuring that each state displays a distinct combination of lights

1.2 Apparatus

Bread Board, 1 7476IC(2JK flip-flops), 1 7408IC(AND gate), 1 7402IC (NOR gate), 3 Green LED′s, 3 Red LED′s, 6 resistors, wires, Arduino/AFG (For clock), Power supply(In case of AFG)

1.3 Theory

1. State Transition Table

The system has three valid states (00, 01, 10) representing the three states (S_0, S_1, and S_2). If the system enters the invalid state (1,1), it is forced to return to (0,0). We use the JK flip-flop excitation table to determine the inputs J_1, K_1, J_0, K_0.

Present State		Flip-Flop 1		Flip-Flop 0		Next State	
Q_1	Q_0	J_1	K_1	J_0	K_0	Q_1	Q_0
0	0	0	X	1	X	0	1
0	1	1	X	X	1	1	0
1	0	X	1	0	X	0	0
1	1	X	1	X	1	0	0

JK Flip-Flop Logic Equations

By simplifying the excitation columns (treating X as 1 where helpful):

- $J_1 = Q_0$
- $K_1 = 1$
- $J_0 = \overline{Q_1}$
- $K_0 = 1$

LED Output Truth Table

We treat the outputs for the invalid state (1,1) as Don't Cares (X) to simplify the logic for the Green (G) and Red (R) LEDs.

Q_1	Q_0	G_0	R_0	G_1	R_1	G_2	R_2
0	0	1	0	0	1	0	1
0	1	0	1	1	0	0	1
1	0	0	1	0	1	1	0
1	1	X	X	X	X	X	X

Karnaugh Maps and Final Simplified Logic

State 0 Logic:

Green 0 (G_0):

$Q_1 \setminus Q_0$	0	1
0	1	0
1	0	X

$\Rightarrow G_0 = \overline{Q_1} \cdot \overline{Q_0}$

Red 0 (R_0):

$Q_1 \setminus Q_0$	0	1
0	0	1
1	1	X

$\Rightarrow R_0 = Q_1 + Q_0$

State 1 Logic:

Green 1 (G_1):

$Q_1 \setminus Q_0$	0	1
0	0	1
1	0	X

Grouping (0,1) and (1,1) $\Rightarrow G_1 = Q_0$

Red 1 (R_1):

$Q_1 \setminus Q_0$	0	1
0	1	0
1	1	X

Grouping (0,0) and (1,0) $\Rightarrow R_1 = \overline{Q_0}$

State 2 Logic:

Green 2 (G_2):

$Q_1 \setminus Q_0$	0	1
0	0	0
1	1	X

Grouping (1,0) and (1,1) $\Rightarrow G_2 = Q_1$

Red 2 (R_2):

$Q_1 \setminus Q_0$	0	1
0	1	1
1	0	X

Grouping (0,0) and (0,1) $\Rightarrow R_2 = \overline{Q_1}$

Final Logic

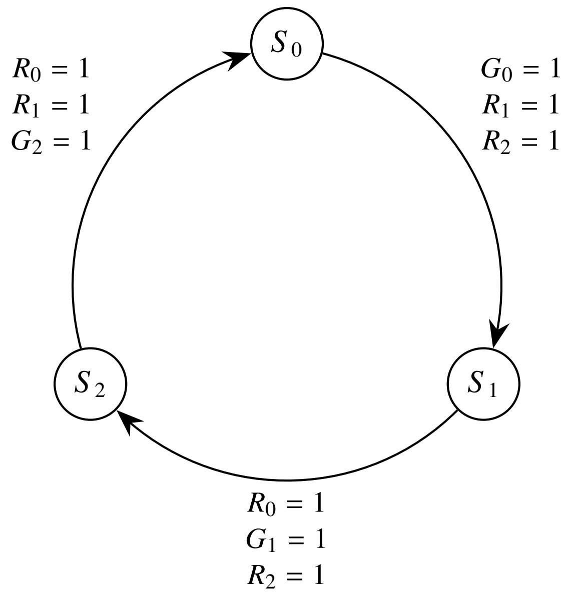
JK Flip-Flop Logic

- $J_0 = \overline{Q_1}$
- $K_0 = 1$
- $J_1 = Q_0$
- $K_1 = 1$

LED Logic

- $G_0 = \overline{Q_1} \cdot \overline{Q_0}$
- $R_0 = Q_1 + Q_0$
- $G_1 = Q_0$
- $R_1 = \overline{Q_0}$
- $G_2 = Q_1$
- $R_2 = \overline{Q_1}$

State Transition Diagram



1.4 Connections

- This is the most important part in the experiment as even a single faulty connection will lead to wrong output.
- The circuit requires three IC′S. We will use one 74HC76 IC (Dual JK Flip-Flops).
- Let IC1 contain Q_0 and Q_1.
- We also use a 74HC08 (Quad AND) as IC2 and a 74HC02 (Quad NOR) as IC3.

Step 1: Power, Ground

- **VCC (+5V):** Connect to IC1 Pin 5, IC2 Pin 14, and IC3 Pin 14.
- **Ground (0V):** Connect to IC1 Pin 13, IC3 Pin 7, and IC4 Pin 7.
- **PRESET and SET:** The 74HC76 has active-low Preset and Set pins. Tie them all to HIGH (+5V) : IC1 Pins 2, 3, 7, 8 .

Step 2: Synchronous Clock

- Connect the output of the 1.00 Hz Function Generator simultaneously to the clock inputs of flip-flop: IC1 Pin 1 .

Step 3: LSB Flip-Flop (Q_0) Wiring

- $J_0 = \overline{Q_1}$**:** Connect IC1 Pin 10 ($\overline{Q_1}$) directly to IC1 Pin 4 (J_0).
- $K_0 = 1$**:** Connect IC1 Pin 12 (K_0) directly to +5V.

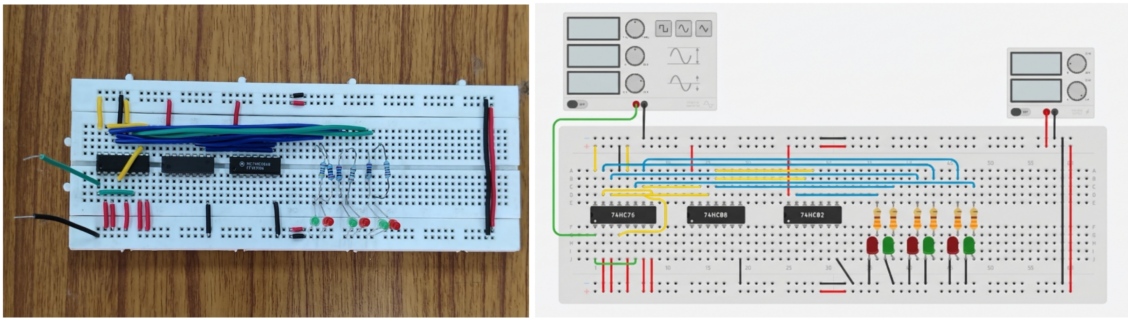
Step 4: MSB Flip-Flop (Q_1) Wiring

- $J_1 = Q_0$**:** Connect IC1 Pin 15 (Q_0) directly to IC1 Pin 9 (J_1).
- $K_1 = 1$**:** Connect IC1 Pin 16 (which is K_1) directly to +5V.

Step 5: LED′S Wiring

- $G_0 = \overline{Q_1} \cdot \overline{Q_0}$
  - Connect IC1 pin 14($\overline{Q_0}$) to IC2 pin 13
  - Connect IC1 pin 10($\overline{Q_1}$) to IC2 pin 12
  - Connect IC2 pin 11 to the left most green LED
- $R_0 = Q_1 + Q_0$
  - Use the fact that $Q_1 + Q_0 = \overline{\overline{Q_1} \cdot \overline{Q_0}}$
  - Connect IC1 pin 14($\overline{Q_0}$) to IC2 pin 13
  - Connect IC1 pin 10($\overline{Q_0}$) to IC2 pin 12
  - Connect IC2 pin 11 to the left most red LED
- $G_1 = Q_0$**:** Connect IC1 pin 15(Q_0) to the middle green LED
- $R_1 = \overline{Q_0}$**:** Connect IC1 pin 14($\overline{Q_0}$) to the middle red LED
- $G_2 = Q_1$**:** Connect IC1 pin 11(Q_1) to the right most green LED
- $R_2 = \overline{Q_1}$**:** Connect IC1 pin 10($\overline{Q_1}$) to the right most red LED

1.5 Circuit Diagram



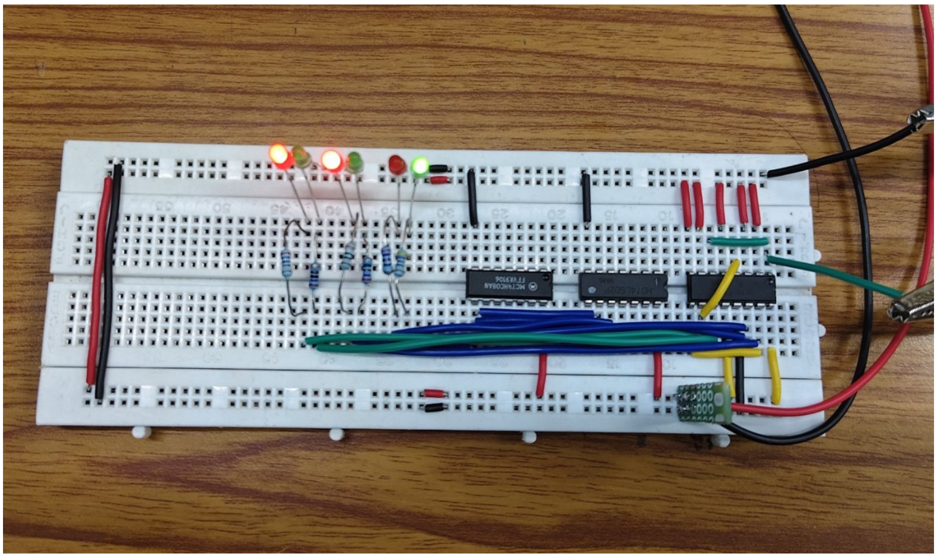
Experimental and Simulated circuit diagram

1.6 Observations

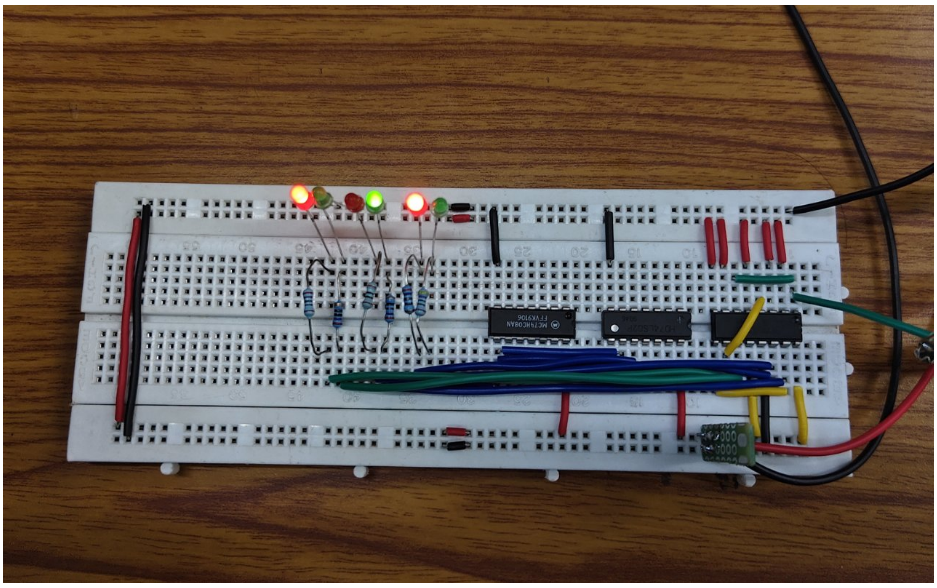
- Clock which updates the value of the counter can be given in two ways using arduino and using AFG.
- Here we have given clock using AFG where we set the time period to 60s and 50% duty cycle
- Initially we connected the left most green led connections to middle one and vice versa then we got a random state of the led′s
- Then we started debugging the circuit from the scratch then we found the error and we fixed it and we got the output in correct order

Note:
The photos of output captured are taken from the other side of the circuit i.e in state S_0 left green led and the other 2 red led′s should glow but in the photo you will see right green led and the other 2 red led′s lighting as the photo is captured in reverse angle

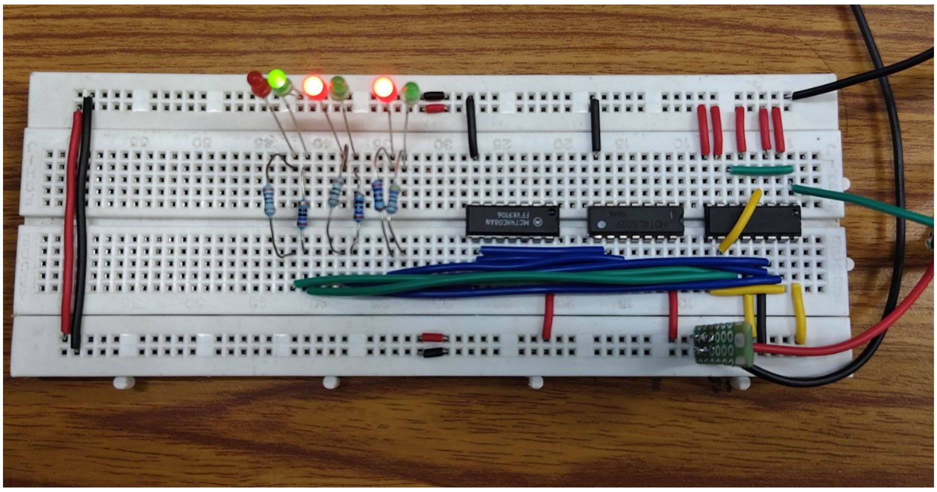
1.7 Output Diagrams



Output for S_0 ($G_0 = 1$ $R_1 = 1$ $R_2 = 1$)



Output for S_1 ($G_1 = 1$ $R_0 = 1$ $R_2 = 1$)



Output for S_2 ($G_2 = 1\ R_0 = 1\ R_1 = 1$)

1.8 Pin Diagram of IC′S Used